Secret-Learn — Privacy-preserving ML (191 algorithms × FL / SL / SS)

github.com/chenxingqiang/Secret-Learn · sklearn API · JAX acceleration · SecretFlow devices

1. Applications

- **Healthcare** — hospitals keep records local
- **Finance** — cross-bank fraud / risk
- **IoT / edge** — federated sensors
- **Research** — multi-institution studies

2. sklearn-compatible API (191 estimators, 30+ categories)

- Linear / GLM
- Ensemble / trees
- SVM / neighbors
- Cluster / PCA
- NN / Naive Bayes
- Preprocess / select

3. Privacy mode router

- **FL — Federated**
  data stays on PYU
  ~~local JAX train~~
  HEU secure aggregate
  best: horizontal FL
- **SL — Split learning**
  model split by party
  ~~local JAX~~
  encrypted activations
  vertical / deep models
- **SS — Secret sharing**
  SPU MPC encrypt
  maximum privacy
  multi-party runtime

4. Intelligent generation + acceleration

- **Algorithm classifier**
  supervised? partial_fit?
  fit(x) vs fit(x,y)
- **Template migrator**
  auto-generate FL/SL/SS
  adapter from sklearn
- **JAX-sklearn core**
  local 5×+ speedup
  CPU / GPU / TPU
- **573 examples**
  one per algo × mode
  batch runners

5. SecretFlow device plane

- **PYU**
  plaintext local
  party compute
- **HEU**
  homomorphic
  secure aggregate
- **SPU**
  MPC / secret share
- **TEE**
  trusted execution
- **DP hooks**
  optional noise

Same estimator call; mode picks FL/SL/SS. FL keeps data local, SL splits the net, SS runs under MPC on SPU.